

System Architecture Document

Multi-Tenant Hierarchical Workflow and Form-Based Verification Platform

1. Overview

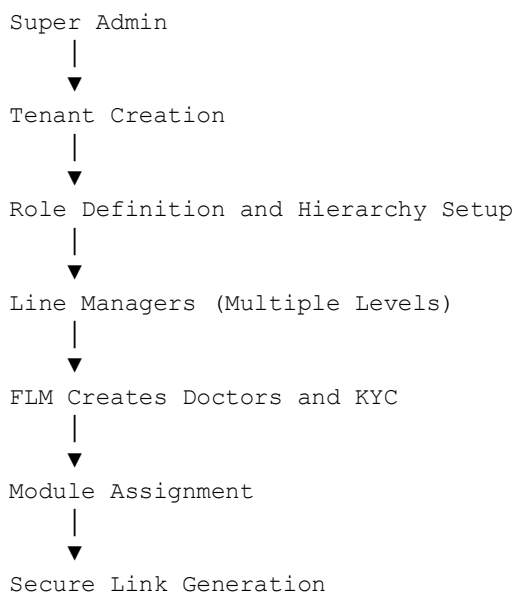
This document describes the architecture of a multi-tenant platform designed to manage hierarchical user roles, dynamic module execution, doctor onboarding, KYC validation, and document-based approval workflows. The system supports flexible organizational structures with unlimited hierarchy levels and a secure, link-based interaction model for external users.

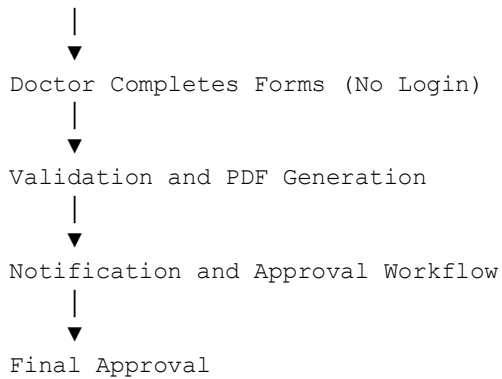
2. Core Components

The platform consists of the following core components:

- Multi-Tenant Management
- Dynamic Role and Hierarchy Engine
- Doctor and KYC Management
- Dynamic Module/Form Builder
- Token-Based Access System
- PDF Generation and Document Aggregation
- Notification and Approval Workflow Engine

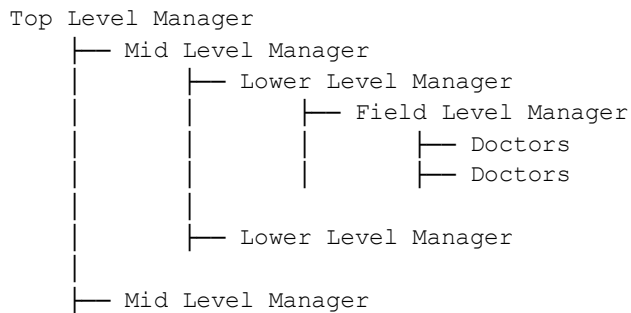
3. High-Level Architecture Flow





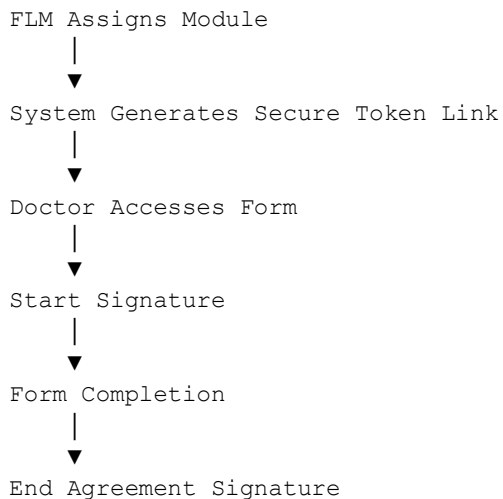
4. Multi-Tenant and Hierarchy Model

Each tenant operates in an isolated environment. Users are organized in a hierarchical structure using a parent-child relationship model. The hierarchy supports unlimited levels, allowing dynamic expansion based on business requirements.



5. Doctor Interaction Model (No Authentication)

Doctors interact with the system through secure, token-based links delivered via email or messaging platforms. No login is required. Each link is uniquely generated and mapped to a specific module instance.



↓
Submission

6. Module Execution Flow

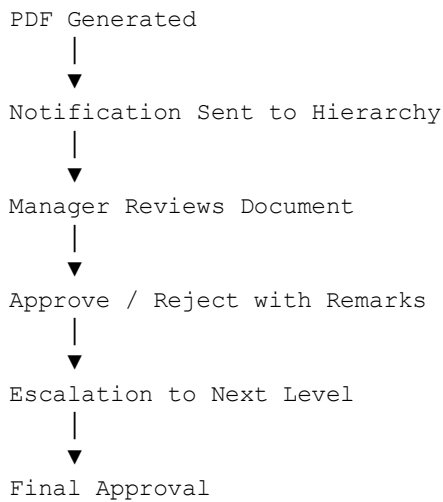
Module Start
↓
Mandatory Start Signature
↓
Dynamic Form Rendering (MCQ / Custom Fields)
↓
Response Submission
↓
Mandatory End Signature
↓
Completion Status Updated

7. PDF Generation and Document Aggregation

Upon completion of all required activities, the system generates a consolidated PDF document containing all relevant data for verification.

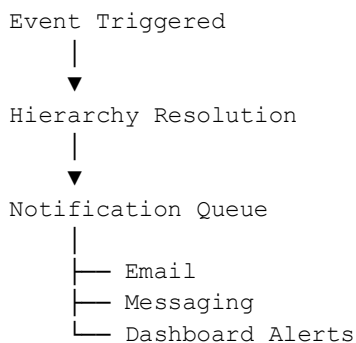
Doctor Completes All Requirements
↓
System Validation
↓
Data Aggregation (Profile, KYC, Modules, Signatures)
↓
HTML Template Rendering
↓
PDF Generation
↓
Storage and Link Creation

8. Approval Workflow



9. Notification System

The system includes an event-driven notification mechanism. Notifications are triggered based on system events and delivered via multiple channels.



10. Data Transfer and Role Changes

The system supports dynamic reassignment of users and data in cases of role changes, promotions, or exits. All associated data such as doctors and subordinate users can be reassigned without data loss.

11. Security and Access Control

- Token-based access for external users
- Role-based access control for internal users
- Secure storage for documents and generated PDFs
- Audit tracking for all major actions

12. Technology Stack (Recommended)

- Frontend: React.js
- Backend: Node.js with Express.js
- Database: MongoDB (or AWS DocumentDB based on client preference)
- Hosting & Infrastructure: Amazon Web Services (AWS)
 - Compute: EC2 / Lambda
 - Storage: S3 (for documents and PDFs)
 - Email Services: AWS SES
 - Queue/Processing: AWS SQS / SNS (or Redis if required)
- PDF Generation: HTML to PDF rendering engine
- Form setup : need to be explored based on requirements

Note: The technology stack can be adjusted based on client requirements during development.